Test on VMMV 5.5.6

Test results

Status: PASSED

Status of tested combinations:

- VM from page load to Rendered glasses
  - Mid-end mobile device: Passed
- VM slow horizontal head movement
  - Mid-end mobile device: Passed
- VM from e-commerce page load to rendered "Try" button
  - Mid-end mobile device: Passed
- VMP video processing phase
  - Mid-end mobile device: Passed
- VMP slow horizontal head movement
  - Mid-end mobile device: Passed
- VMP picture processing phase
  - Mid-end mobile device: Passed
- Multi-VM all glasses rendered
  - Mid-end mobile device: Failed (Actionable Items)
- Multi-VM slow horizontal head movement
  - Mid-end mobile device: Failed (Actionable Items)

Report

Performance tests on VMMV 5.5.6 **passed.**

Analysis compared VMMV 5.5.6 with the baseline VMMV 5.5.2.

Results confirm stable performance KPIs for VM and VMP in all tested cases, with improved main thread computation during VM engine initialization and enhanced VMP picture resource handling.

No UI/UX issues occurred during testing.

Multi-VM insight:

VMMV 5.5.6 enhances Multi-VM rendering by moving heavy logic to a web worker, which reduces the longest main-thread task duration.
However, important regressions persist:

- Multi-VM runtime processing enforces the computation limit as a single VM nears it
- INP score remains high when clicking the MultiVM toggle.

These points need to be addressed before enabling the MultiVM feature on e-commerce.

Last measurement: Jan 26, 2026

List of tested combinations

VM from page load to Rendered glasses

Mid-end mobile device

- **Status**: Passed
- **Tested combination**:
  - VMMV 5.5.2 vs VMMV 5.5.6 (TV20 vs TV25_01 - TC01 - TD31_02)
- **Detailed scores**: Detailed scores
- **Profiling data**: Profiling data

Report
Analysis confirms all performance KPIs remain stable without significant degradation, with an important reduction of Longest task duration on the main thread due to relocating heavy logic to a web worker.

KPI	VMMV 5.5.2	VMMV 5.5.6	Δ	%	Threshold
Rendering time (s)	7.20	6.65	-0.55	-8%	-
Scripting (%)	43%	52%	9.77%	23%	80%
Long tasks > 100 ms	7	5	-2	-24%	-

Long tasks > 500 ms	0	0	0	N/A	-
Longest task (ms)	380	186	-194.26	-51%	1500
JS heap min (MB)	12.98	12.98	0.00	0%	30
JS heap max (MB)	51.47	50.40	-1.07	-2%	80
JS heap min/max Δ (MB)	38.49	37.42	-1.07	-3%	30
Network Requests	68	68	0.00	0%	-
Network MB transferred	9.60	9.40	-0.20	-2%	-
Network MB resources	14.20	13.30	-0.90	-6%	-

Back on top

VM slow horizontal head movement

Mid-end mobile device

- **Status**: Passed
- **Tested combination**:
  - VMMV 5.5.2 vs VMMV 5.5.6 (TV20 vs TV25_01 - TC04 - TD31_02)
- **Detailed scores**: Detailed scores
- **Profiling data**: Profiling data

Report

Analysis confirms all performance KPIs remain stable without significant degradation.

KPI	VMMV 5.5.2	VMMV 5.5.6	Δ	%	Threshold
Rendering time (s)	1.01	1.02	0.01	1%	-
Scripting (%)	96%	98%	1.76%	2%	80%
Long tasks > 100 ms	1	2	1	150%	-
Long tasks > 500 ms	0	0	0	N/A	-
Longest task (ms)	97	110	13.25	14%	1500
JS heap min (MB)	25.62	24.43	-1.19	-5%	30
JS heap max (MB)	33.43	32.17	-1.27	-4%	80
JS heap min/max Δ (MB)	7.81	7.73	-0.08	-1%	30
FPS (estimate)	11	11	0.00	0%	15
FPS (total)	60	60	0.00	0%	-
Longest frame (ms)	16.60	16.70	0.10	1%	-
Complete Frames (%)	18.33%	18.33%	0.00%	0%	-

Partial Presented Frames (%)	81.67%	81.67%	0.00%	0%	-
Idle Frames (%)	0.00%	0.00%	0.00%	N/A	-
Dropped Frames (%)	0.00%	0.00%	0.00%	N/A	-

Back on top

VM from e-commerce page load to rendered "Try" button

Mid-end mobile device

- **Status**: Passed
- **Tested combination**:
  - VMMV 5.5.2 vs VMMV 5.5.6 (TV20 vs TV25_01 - TC12 - TD28)
- **Detailed scores**: Detailed scores
- **Profiling data**: Profiling data

Report

Integration test performed using Chrome override to include the compared versions in this PDP:
https://www.oakley.com/en-us/product/W0OO9406?variant=888392404756
Analysis confirms all performance KPIs remain stable without significant degradation.

KPI	VMMV 5.5.2	VMMV 5.5.6	Δ	%	Threshold
Network Requests	15	16	1.00	7%	-
Network MB	0.20	0.20	0.00	-1%	-

transferred					
Network MB resources	0.74	0.72	-0.02	-3%	-

Back on top

VMP video processing phase

Mid-end mobile device

- **Status**: Passed
- **Tested combination**:
  - VMMV 5.5.2 vs VMMV 5.5.6 (TV20 vs TV25_01 - TC10 - TD31_02)
- **Detailed scores**: Detailed scores
- **Profiling data**: Profiling data

Report

Analysis confirms all performance KPIs remain stable without significant degradation.

Even if confirmed, the current video processing time is too long for a smooth user experience and should be addressed as soon as possible.

KPI	VMMV 5.5.2	VMMV 5.5.6	Δ	%	Threshold
Rendering time (s)	35.51	32.82	-2.69	-8%	-
Scripting (%)	57%	59%	1.40%	2%	80%
Long tasks > 100 ms	2	5	3	150%	-
Long tasks > 500 ms	1	1	0	33%	-

Longest task (ms)	624	642	18.00	3%	1500
JS heap min (MB)	23.85	21.82	-2.03	-9%	30
JS heap max (MB)	52.50	55.82	3.32	6%	80
JS heap min/max Δ (MB)	28.65	34.00	5.35	19%	30
Network Requests	67	66	-1.00	-1%	-
Network MB transferred	12.40	12.10	-0.30	-2%	-
Network MB resources	35.00	33.40	-1.60	-5%	-

KPI	VMMV 5.5.2 (no profiling)	VMMV 5.5.6 (no profiling)	Δ	%	Threshold
Rendering time (s)	29.33	29.92	0.59	2%	-

Back on top

VMP slow horizontal head movement

Mid-end mobile device

- **Status**: Passed
- **Tested combination**:
  - VMMV 5.5.2 vs VMMV 5.5.6 (TV20 vs TV25_01 - TC11 - TD31_02)
- **Detailed scores**: Detailed scores
- **Profiling data**: Profiling data

Report

Analysis confirms all performance KPIs remain stable without significant degradation.

KPI	VMMV 5.5.2	VMMV 5.5.6	Δ	%	Threshold
Rendering time (s)	1.00	1.00	0.00	0%	-
Scripting (%)	19%	20%	1.54%	8%	80%
Long tasks > 100 ms	0	0	0	N/A	-
Long tasks > 500 ms	0	0	0	N/A	-
Longest task (ms)	7	10	2.99	41%	1500
JS heap min (MB)	54.30	40.73	-13.57	-25%	30
JS heap max (MB)	59.53	44.37	-15.17	-25%	80
JS heap min/max Δ (MB)	5.23	3.63	-1.60	-31%	30
FPS (estimate)	59	56	-3.33	-6%	15

FPS (total)	60	59	-0.67	-1%	-
Longest frame (ms)	16.73	27.80	11.07	66%	-
Complete Frames (%)	98.89%	94.38%	-4.51%	-5%	-
Partial Presented Frames (%)	0.00%	0.00%	0.00%	N/A	-
Idle Frames (%)	0.00%	0.00%	0.00%	N/A	-
Dropped Frames (%)	0.00%	5.62%	5.62%	N/A	-

Back on top

VMP picture processing phase

Mid-end mobile device

- **Status**: Passed
- **Tested combination**:
  - VMMV 5.5.2 vs VMMV 5.5.6 (TV20 vs TV25_01 - TC13 - TD31_02)
- **Detailed scores**: Detailed scores
- **Profiling data**: Profiling data

Report

VMMV 5.5.6 improves performance by skipping the unnecessary download of the ffmpeg and TensorFlow libraries during the VMP picture phase and reducing the hardcoded loop for picture pose inference, freeing the main thread.

KPI	VMMV 5.5.2	VMMV 5.5.6	Δ	%	Threshold
Rendering time (s)	5.20	2.81	-2.39	-46%	-
Scripting (%)	53%	60%	7.24%	14%	80%
Long tasks > 100 ms	4	5	1	25%	-
Long tasks > 500 ms	1	1	0	0%	-
Longest task (ms)	1553	619	-934.37	-60%	1500
JS heap min (MB)	22.09	17.62	-4.47	-20%	30
JS heap max (MB)	53.11	36.22	-16.89	-32%	80
JS heap min/max Δ (MB)	31.02	18.60	-12.42	-40%	30
Network Requests	89	56	-33.00	-37%	-
Network MB transferred	12.40	1.80	-10.60	-85%	-
Network MB resources	37.93	1.00	-36.93	-97%	-
INP	297	297	0.00	0%	200

Back on top

Multi-VM all glasses rendered

Mid-end mobile device

- **Status**: Failed
- **Tested combination**:
  - VMMV 5.5.2 vs VMMV 5.5.6 (TV21 vs TV25_02 - TC05 - TD31_02)
- **Detailed scores**: Detailed scores
- **Profiling data**: Profiling data

Report

VMMV 5.5.6 improves Multi-VM rendering by significantly reducing the longest task duration on the main thread through relocating heavy logic to a web worker.

The high INP score remains unaddressed, and network resources have increased (this increase was not present in the previous version VMMV 5.5.4).

KPI	VMMV 5.5.2	VMMV 5.5.6	Δ	%	Threshold
Rendering time (s)	2.76	3.64	0.88	32%	-
Scripting (%)	87%	71%	-15.80%	-18%	80%
Long tasks > 100 ms	5	7	1	25%	-
Long tasks > 500 ms	1	0	-1	-100%	-
Longest task (ms)	598	386	-212.28	-35%	1500
JS heap min (MB)	26.37	27.08	0.71	3%	30
JS heap max (MB)	58.25	67.00	8.75	15%	80

JS heap min/max Δ (MB)	31.88	39.92	8.04	25%	30
Network Requests	41	28	-13.00	-32%	-
Network MB transferred	2.10	5.30	3.20	152%	-
Network MB resources	0.53	5.80	5.27	994%	-
INP click to close	410	379	-31.00	-8%	200
INP click to open	321	356	35.44	11%	-

Actionable Items

Multi-VM | Improve INP on open/close click

Back on top

Multi-VM slow horizontal head movement

Mid-end mobile device

- **Status**: Failed
- **Tested combination**:
  - VMMV 5.5.2 vs VMMV 5.5.6 (TV24 vs TV22 - TC04 vs TC06 - TD31)
- **Detailed scores**: Detailed scores
- **Profiling data**: Profiling data

Report

VMMV 5.5.6 Multi-VM shows KPIs similar to VMMV 5.5.6 VM but enforces the computation limit

since a single VM nears it.

KPI	VMMV 5.5.4 (Multi-VM OFF)	VMMV 5.5.4 (Multi-VM ON)	Δ	%	Threshold
Rendering time (s)	1.02	1.01	-0.01	-1%	-
Scripting (%)	98%	97%	-1.11%	-1%	80%
Long tasks > 100 ms	2	7	6	340%	-
Long tasks > 500 ms	0	0	0	N/A	-
Longest task (ms)	110	125	14.33	13%	1500
JS heap min (MB)	24.43	30.11	5.68	23%	30
JS heap max (MB)	32.17	41.57	9.41	29%	80
JS heap min/max Δ (MB)	7.73	11.46	3.73	48%	30
FPS (estimate)	11	9	-2.33	-21%	15
FPS (total)	60	60	0.00	0%	-
Longest frame (ms)	16.70	16.60	-0.10	-1%	-
Complete Frames (%)	18.33%	14.44%	-3.89%	-21%	-

Partial Presented Frames (%)	81.67%	85.56%	3.89%	5%	-
Idle Frames (%)	0.00%	0.00%	0.00%	N/A	-
Dropped Frames (%)	0.00%	0.00%	0.00%	N/A	-

Actionable Items

VM CORE | Analysis of pose inference and rendering strategy on runtime execution

Back on top